COMP 2030R Lab 2

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1. MIPS has two groups of data transfer instructions: 'load' to store data in a register and 'store' to save register data in a memory location.

   'load' has four variations:

```
li $t0, 3          # load immediate: store a constant in a register
la $t0, A          # load address: store an address (label) in a register
lw $t0, A($t9)     # load word: fetch a word (4 bytes) from memory to store in a register
lb $t0, A($t9)     # load word: fetch a single byte from memory to store in a register
```

   Given the three integers in the data segment, consider the program below. In the comment, write the updated value of the destination register when the instruction completes its execution. Assume the data segment starts at address 0x1100.

```
   .data
array:   .word            0x37, 0x55, 0xF

   .text
   la    $s0, array        # $s0 = 0x1100

   la    $s1, array+4      # $s1 = 0x1104

   la    $s2, 8($s0)       # $s2 = 0x1108

   li    $a0, 4            # $a0 = 4

   lw    $t0, 0($s0)       # $t0 = 0x37

   lw    $t3, array($a0)   # $t3 = 0x55 in big endian

   lw    $t1, 8($s0)       # $t1 = 0xF in big endian

   lb    $t2, ($s0)        # $t2 = 0x37 in big endian
```

2. Suppose you have a 10-element array 'arr' with integers (words), declared with initial values of 0:

```
        .data
arr:    .word       0:10
```

   a. Write MIPS instructions to store $t0 and $t1 to the first and second elements of arr.

      sw $t0, (arr)

      sw $t1, 4(arr)

   b. Write MIPS instructions to add the 2nd and 3rd elements of arr and save the sum in the 4th element.

      sw $t2, 8(arr)
      add $t3, $t1, $t2
      sw $t3, 16(arr)

3. Suppose that all numbers are limited to 6-bit binary quantities. In parentheses, write decimal values of the given binary numbers.
   a. x=001011 (11)
      Left shift x by one bit, and write the resulting binary bit sequence and its decimal equivalent in the parenthesis.

      y=__010110_____ (22)

   b. Let's say that the binary sequence in part (a) is called 'y.'
      Left shift y by one bit, and write the resulting binary bit sequence and its decimal equivalent in the parenthesis.

      __101100_____ (44)

   c. z=011010 ( )
      Right shift z by one bit, and write the resulting binary bit sequence and its decimal equivalent in the parenthesis.

      ___001101_____ (13)

   d. The left shift of x by one bit is writtein in C as
      `y = x << 1;`
      What arithmetic operation is equivalent to a left shift by one bit? Rewrite the statement y=x<<1 using only arithmetic operations (+, -, *, or /):

      `y = __x * 2_____ ;`